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MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT, M9 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Catapult, Aircraft Ejection Seat, M9, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons, General Specification for
MIL–I–6866 —Inspection, Penetrant, Method of
MIL–I–6868 —Inspection, Process, Magnetic Particle
MIL–I–45208 —Inspection System Requirements

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes
MIL–STD–109 —Quality Assurance Terms and Definitions
MIL–STD–413 —Visual Inspection Guide for Rubber O-Rings
MIL–STD–639 —Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

MUNITIONS COMMAND

F8596883 —Catapult, Aircraft Ejection Seat, M9, Metal Parts Assembly
C8826679 —Container, Ammunition, Fiber, M424 for Catapult, M9

MUNITIONS COMMAND

C8826680 —Box, Packing, Ammunition, for Catapult, M9 in Fiber Container, M424
IEL–8596883 —Index of Inspection Equipment List for Catapult, Aircraft Ejection Seat, M9, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **General.** Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

FSC 1375

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3.2 **Initial production sample.** Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 **Material.** The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.4 **Physical properties.** Physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 **Hydrostatic pressure.** The components listed in table I shall withstand the corresponding internal pressure for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

TABLE I

Component	*Hydrostatic Pressure (psi) (min.)*
Tube, Motor	3,500
Tube, Launcher	1,200
Cylinder, Slider Valve and Tube, Booster Assembly	1,200
Can, Bottom	1,200

3.5.1 The breech, launcher shall withstand an internal pressure of 1,200 psi, minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1). The pressure shall be confined to an area such that no pressure is applied to the bottom surface of the breech.

3.6 **Air pressure.** The nozzle cups in nozzle assembly shall withstand an internal pressure of 100 psi, minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1). Pressure shall be applied to inlet side of nozzle.

3.7 **Tensile load.** The nozzle assembly shall withstand a tensile load of 4,000 pounds, minimum, applied between retainer tang and nozzle along the horizontal axis of the nozzle assembly without separation or deformation of the components.

3.8 **Shearpin strength.** The pin shall shear within 41 pounds plus or minus 5 pounds upon application of a load at the rate of 0.125 inches per minute.

3.9 **Valve shearpin strength.** The valve shearpin shall shear within 164 pounds plus or minus 50 pounds upon application of a load at the rate of 0.125 inches per minute.

3.10 **Structural strength.** The metal parts assembly shall withstand the test loads of 4,000 pounds, minimum, in tension and 8,000 pounds, minimum, in compression applied at the mounting points for 15 seconds, minimum, without mechanical failure (see 6.1).

3.11 **Protective finish.** Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.12 **Residual magnetism.** The metal parts asembly shall not deflect the indicator of the compass more than 3° in either direction when passed in front of the compass at a distance of 6 inches.

3.13 **Workmanship.** The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.13.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.13.2 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.13.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for

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the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to standard MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 Initial production sample. Prior to the initiation of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 10 sets of parts, 50 shearpins and 50 valve shearpins from the same lot of shearpins tested by the contractor. The samples shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4, except as provided in Specification MIL–I–45208. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Presentation of lots.* The product shall be presented in accordance with Standard MIL–STD–105.

4.3.1.2 *Size of lot.* The shipping lots of metal parts assemblies shall consist of 300 catapults, maximum, exclusive of test samples, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with Standard MIL–STD–413. The Ordnance Standard Finish Blocks prescribed in Standard MIL–STD–639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

		Accept numbers			
		Major		Minor	
Lot size	*Sample size*	*Accept*	*Reject*	*Accept*	*Reject*
1– 300	100% Examination				
301– 1,300	110	1	2	2	3
1,301– 3,200	150	2	3	3	4
3,201– 8,000	225	3	4	4	5
8,001–22,000	300	4	5	5	6

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TABLE III. *Total Allowable Defects per Sample Size*

Sample Size for Major Defects	110	150	225	300
No. of Defects in Class:				
2– 3	1	2	3	4
4– 7	2	3	4	5
8–19	3	4	5	7
20–43	4	5	8	10
over–43	6	8	11	14

Sample Size for Minor Defects	110	150	225	300
No. of Defects in Class:				
2– 3	2	3	4	5
4– 7	3	4	5	7
8–16	4	5	8	10
17–48	6	8	11	14
over–48	8	11	17	20

4.3.2.1 *Spring, Can, Dwg. D8596889, covering a detail of Dwg. F8596883.*

Defect	*Method of inspection*
CRITICAL:	
None defined.	
MAJOR:	
101. Load at specified height	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR:	
201. Diameter	Gage
202. Profile of end	Visual
203. Radius at small end	Visual
204. Angular location of end	Visual
205. Surface finish, improper	Visual

4.3.2.2 *Holder, Cable, Lower, Dwg. C8596890, covering a detail of Dwg. F8596883.*

Defect	*Method of inspection*
CRITICAL:	
None defined.	
MAJOR:	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Symmetry of slot with cavity	Gage
104. Diameter of cavity, max.	Gage
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, max.	Gage
202. Diameter of cavity, min.	Gage
203. Diameter of undercut, min.	Gage
204. Depth of cavity	Gage
205. Depth of slot	Gage
206. Length to shoulder	Gage
207. Width of slot	Gage
208. Angle at shoulder	Gage
209. Diameter of shoulder	Measure
210. Diameter of flat on face	Measure
211. Total length	Measure
212. Width of undercut, max.	Measure
213. Marking missing, incorrect or illegible	Visual
214. Radius or chamfer missing or incorrect	Visual
215. Surface finish, improper	Visual

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4.3.2.3 *Cylinder, Slider Valve, Dwg. F8596894, covering a detail of Dwg. F8596893 (before application of protective finish).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of counterbore, small end with cavity	Gage
102. Diameter of large counterbore, small end, max.	Gage
MINOR:	
201. Diameter of large counterbore, small end, min.	Gage
202. Diameter of middle cavity, small end	Gage
203. Depth of large counterbore, small end	Gage
204. Depth of middle cavity, small end	Gage
205. Total length	Gage

4.3.2.4 *Cylinder, Slider Valve, Dwg. F8596894, covering a detail of Dwg. F8596893 (after application of protective finish).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of cavity with large body adjacent to flange	Gage
102. Concentricity of small counterbore in head with cavity	Gage
103. Concentricity of middle counterbore in head with body adjacent to flange	Gage
104. Concentricity of groove in middle counterbore with counterbore	Gage
105. Concentricity of small groove on head with large body adjacent to flange	Gage
106. True position of hole with cavity	Gage
107. Diameter of long cavity	Gage
108. Diameter of small counterbore, head end	Gage
109. Diameter of pinhole, max.	Gage
110. Diameter of middle counterbore, head end, max.	Gage
111. Diameter of small groove on large body, min.	Gage
112. Diameter of groove in middle counterbore, min.	Gage
113. Diameter of body adjacent to flange, min.	Gage
114. Depth to base of middle counterbore	Gage
115. Depth of small counterbore at cavity, head end	Gage
116. Width at base of groove in middle counterbore, head end	Gage
117. Thickness of wall, min.	Gage
118. Location of groove in middle counterbore from base of middle counterbore	Gage
119. Angle of groove in middle counterbore	Gage
120. Metal defective	Visual
121. Protective finish inadequate or defective	Visual
122. Surface finish of cavity, improper	Visual
123. Burr or sharp corner	Visual
124. Foreign matter	Visual
MINOR:	
201. Diameter of small body	Gage
202. Diameter of pinhole, min.	Gage
203. Diameter of middle counterbore, head end, min.	Gage
204. Diameter of groove in middle counterbore, head end, max.	Gage
205. Diameter of large body adjacent to flange, max.	Gage
206. Diameter of flange	Gage
207. Diameter of small groove on large body, max.	Gage
208. Depth of groove on small body	Gage
209. Length of head	Gage
210. Width of groove on small body	Gage
211. Thickness of flange	Gage
212. Location of pinhole	Gage

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Defect	Method of inspection
213. Location of large groove on large body	Gage
214. Location of slot	Gage
215. Diameter of large counterbore, head end	Measure
216. Diameter of large groove on large body	Measure
217. Depth of large counterbore, head end	Measure
218. Length of slot	Measure
219. Width of slot	Measure
220. Width of large groove on large body	Measure
221. Width of small groove on large body	Measure
222. Width of flange adjacent to small groove on large body	Measure
223. Profile of small groove side walls, max.	Measure
224. Radius or chamfer missing or incorrect	Visual
225. Surface finish improper, except as otherwise classified	Visual

4.3.2.5 *Pin, Valve, Shear, Dwg. C8596901, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Diameter, min.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr or sharp corner	Visual
106. Foreign matter	Visual
MINOR:	
201. Diameter, max.	Gage
202. Total length, max.	Gage
203. Surface finish, improper	Visual

4.3.2.6 *Piston Slider Valve, Dwg. F8596902, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of middle groove with body	Gage
104. Concentricity of end grooves with body	Gage
105. Symmetry of thickness of tab with body	Gage
106. True position of threaded hole	Gage
107. Diameter of body	Gage
108. Diameter of end grooves, min.	Gage
109. Diameter of middle groove, min.	Gage
110. Width of middle groove, max.	Gage
111. Thickness of wall, min.	Gage
112. Variation in sidewalls of forward and rear grooves, max.	Gage
113. Metal defective	Visual
114. Protective finish, inadequate or defective	Visual
115. Burr or sharp corner, except as required	Visual
116. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, min.	Gage
202. Diameter of end grooves, max.	Gage
203. Length to taper	Gage
204. Length of body	Gage
205. Width of middle groove, min.	Gage

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Defect	Method of inspection
206. Thickness of flange adjacent to end groove	Gage
207. Thickness of tab	Gage
208. Location of middle groove	Gage
209. Location of forward groove	Gage
210. Location of threaded hole	Measure
211. Diameter of cavity	Measure
212. Depth to end of tab	Measure
213. Total length	Measure
214. Width of tab	Measure
215. Width of forward and rear groove	Measure
216. Location of forward threaded hole	Measure
217. Permissible variation of groove side walls, max.	Measure
218. Radius or chamfer missing or incorrect	Visual
219. Marking missing, incorrect or illegible	Visual
220. Surface finish, improper	Visual

4.3.2.7 *Cap, Igniter Retaining, Dwg. C8596903, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of hole with body	Gage
102. Diameter of body, min.	Gage
103. Diameter of hole, min.	Gage
104. Thickness	Gage
105. Straightness	Gage
106. Metal defective	Visual
107. Protective finish inadequate or defective	Visual
108. Burr or sharp corner	Visual
109. Foreign matter	Visual
MINOR:	
101. Diameter of body, max.	Gage
202. Diameter of hole, max.	Gage
203. Radius or chamfer missing or incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Surface finish, improper	Visual

4.3.2.8 *Tang, Retainer, Dwg. F8596907, covering a detail of Dwg. F8596883 (before application of protective finish).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of body with small bore and tangs at basic diameter	Gage
102. Concentricity of top of tangs with large bore	Gage
103. Diameter of large bore, max.	Gage
104. Diameter of small bore, max.	Gage
105. Thickness of wall, min.	Gage
106. Burr or sharp corner	Visual
MINOR:	
201. Diameter of large bore, min.	Gage
202. Diameter of small bore, max.	Gage
203. Total length	Gage
204. Spacing of tangs incorrect	Gage
205. Length of small bore	Measure
206. Radius or chamfer missing or incorrect	Visual
207. Surface finish, improper	Visual

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4.3.2.9 *Tang, Retainer, Dwg. F8596907, covering a detail of Dwg. F8596883 (after application of protective finish).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Diameter of body	Gage
102. Length of tang at basic diameter including basic angle	Gage
103. Metal defective	Visual
MINOR:	
201. Diameter over top of tangs	Gage
202. Depth of slot on tang	Gage
203. Width of tang	Measure
204. Number of tangs, incorrect	Visual

4.3.2.10 *Pin, Firing, Dwg. D8596909, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of groove with body	Gage
102. Concentricity of striker with body	Gage
103. Symmetry of wrench holes with body	Gage
104. True position of shearpin hole with body	Gage
105. Diameter of striker	Gage
106. Diameter of shearpin hole, max.	Gage
107. Diameter of body, min.	Gage
108. Diameter of groove, min.	Gage
109. Length of striker	Gage
110. Radius at base of striker	Gage
111. Radial profile of striker	Gage
112. Metal defective	Visual
113. Protective finish inadequate or defective	Visual
114. Surface finish of body and striker, improper	Visual
115. Burr or sharp corner, except as required	Visual
116. Foreign matter	Visual
MINOR:	
201. Diameter of wrench holes	Gage
202. Diameter of body, max.	Gage
203. Diameter of groove, max.	Gage
204. Diameter of shearpin hole, min.	Gage
205. Total length	Gage
206. Location of shearpin hole	Gage
207. Permissible variation in groove sidewalls	Measure
208. Depth of holes	Measure
209. Width of groove	Measure
210. Location of groove	Measure
211. Marking missing, incorrect or illegible	Visual
212. Radius or chamfer missing or incorrect, except as otherwise classified	Visual
213. Surface finish, improper, except as otherwise classified	Visual

4.3.2.11 *Pin, Shear, Dwg. C8596910, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Diameter, min.	Gage
102. Total length, min.	Gage

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Defect	Method of inspection
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr or sharp corner	Visual
106. Foreign matter	Visual
MINOR:	
201. Diameter, max.	Gage
202. Total length, max.	Gage
203. Surface finish, improper	Visual
204. Radius or chamfer missing or incorrect	Visual

4.3.2.12 *Sleeve, Firing Pin, Dwg. C8596915, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of inside diameter with body	Gage
102. True position of pinhole with inside diameter	Gage
103. Diameter of body, min.	Gage
104. Diameter inside, max.	Gage
105. Diameter pinhole, max.	Gage
106. Metal defective	Visual
107. Protective finish inadequate or defective	Visual
108. Surface finish on body and inside diameter	Visual
109. Burr	Visual
110. Foreign matter	Visual
MINOR:	
201. Diameter of body, max.	Gage
202. Diameter inside, min.	Gage
203. Diameter of pinhole, min.	Gage
204. Total length	Gage
205. Location of pinhole	Gage
206. Diameter of flange	Measure
207. Depth of slot	Measure
208. Width of slot	Measure
209. Thickness of flange	Measure
210. Radius or chamfer missing or incorrect	Visual
211. Marking missing, incorrect or illegible	Visual
212. Surface finish, improper, except as otherwise classified	Visual

4.3.2.13 *Spring, Grain, Dwg. D8596916, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Load at specified heights	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR:	
201. Diameter, inside, large end	Measure
202. Diameter, inside, small end	Measure
203. Radius missing or incorrect	Visual
204. Surface finish improper	Visual

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4.3.2.14 *Wire, Lock, Dwg. D8596918, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Width, min.	Gage
102. Thickness, min.	Gage
103. Straightness	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr or sharp corner	Visual
107. Foreign matter	Visual
MINOR:	
201. Width, max.	Gage
202. Thickness, max.	Gage
203. Thickness at tang end	Gage
204. Height of tang	Gage
205. Height at chamfer end	Measure
206. Angle of tang	Measure
207. Radius or chamfer missing or incorrect	Visual
208. Surface finish, improper	Visual

4.3.2.15 *Wire, Lock, Dwg. D8596919, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Width, min.	Gage
102. Thickness, min.	Gage
103. Straightness	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr or sharp corner	Visual
107. Foreign matter	Visual
MINOR:	
201. Width, max.	Gage
202. Thickness, max.	Gage
203. Thickness at tang end	Gage
204. Height of tang	Gage
205. Height at chamfer end	Measure
206. Angle of tang	Measure
207. Radius or chamfer missing or incorrect	Visual
208. Surface finish, improper	Visual

4.3.2.16 *Breech, Launcher, Dwg. D8596884, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of pitch diameter of thread with cavity	Gage
104. Diameter of undercut, max.	Gage
105. Diameter of body, min.	Gage
106. Diameter of cavity, max.	Gage
107. Length to base of cavity, max.	Gage
108. Total length, min.	Gage
109. Thickness of wall, min.	Gage

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Defect	Method of inspection
110. Metal defective	Visual
111. Protective finish inadequate or defective	Visual
112. Surface finish of cavity, improper	Visual
113. Burr or sharp corner	Visual
114. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, min.	Gage
202. Minor diameter of thread, min.	Gage
203. Diameter of body, max.	Gage
204. Diameter of cavity, min.	Gage
205. Diameter of wrench holes	Gage
206. Depth to undercut	Gage
207. Depth of cavity, min.	Gage
208. Depth of wrench holes	Gage
209. Total length, max.	Gage
210. Location of wrench holes	Gage
211. Width of undercut	Measure
212. Radius or chamfer missing or incorrect	Visual
213. Marking missing, incorrect or illegible	Visual
214. Surface finish, improper, except as otherwise classified	Visual

4.3.2.17 *Tube, Booster, Dwg. F8596895, covering a detail of Dwg. F8596883 (before application of protective finish).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of small body with body (counterbore end)	Gage
102. Concentricity of counterbore with small body	Gage
103. Diameter of small body (counterbore end), min.	Gage
104. Straightness	Gage
105. Burr or sharp corner	Visual
MINOR:	
201. Diameter of small body (counterbore end), max.	Gage
202. Depth of counterbore	Gage
203. Total length	Gage
204. Length of small body (counterbore end)	Gage
205. Location of flat	Measure
206. Radius or chamfer missing or incorrect	Visual
207. Surface finish, improper	Visual

4.3.2.18 *Tube, Booster, Dwg. F8596895, covering a detail of Dwg. F8596883 (after application of protective finish).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of groove with small body	Gage
102. Diameter of small body (groove end), min.	Gage
103. Diameter of counterbore	Gage
104. Diameter of groove, min.	Gage
105. Thickness of wall, min.	Gage
106. Thickness of wall at flat	Gage
MINOR:	
201. Diameter of small body (groove end), max.	Gage
202. Diameter of groove, max.	Gage
203. Diameter of large body	Gage

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Defect	Method of inspection
204. Diameter inside	Gage
205. Width of groove	Gage
206. Length of small body (groove end)	Measure
207. Length of flat	Measure
208. Location of groove	Measure
209. Profile of groove	Measure

4.3.2.19 *Head, Motor Tube, Dwg. F8596904, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Pitch diameter of setscrew thread, max.	Gage
102. Minor diameter of setscrew thread, max.	Gage
103. Pitch diameter gas intake port thread, max.	Gage
104. Minor diameter gas intake port thread, max.	Gage
105. Concentricity of firing pin cavity with large cavity	Gage
106. Concentricity of groove in firing pin cavity with firing pin cavity	Gage
107. Concentricity of large groove with large cavity	Gage
108. Concentricity of body with cavity	Gage
109. Perpendicularity of setscrew thread with mounting hole	Gage
110. Symmetry of slot with large groove	Gage
111. Symmetry of thickness of rectangular section with cavity	Gage
112. Symmetry of width of rectangular section with cavity	Gage
113. True position of mounting hole with cavity	Gage
114. True position of setscrew thread with cavity	Gage
115. Diameter of large cavity, max.	Gage
116. Diameter of firing pin cavity, max.	Gage
117. Diameter of groove in firing pin cavity, max.	Gage
118. Diameter of counterbore of gas intake port	Gage
119. Diameter of mounting hole	Gage
120. Diameter of chamfer of gas intake port	Gage
121. Diameter of groove in large cavity	Gage
122. Depth to base of counterbore at gas intake port	Gage
123. Depth to end of effective thread in gas port, min.	Gage
124. Width of large groove, max.	Gage
125. Angle of chamfer at entrance of gas inlet port	Gage
126. Metal defective	Visual
127. Protective finish inadequate or defective	Visual
128. Surface finish of chamfers on gas intake port mouth, improper	Visual
129. Burr or sharp corner	Visual
130. Foreign matter	Visual
MINOR:	
201. Pitch diameter of setscrew thread, min.	Gage
202. Pitch diameter of gas intake port thread, min.	Gage
203. Diameter of firing pin cavity, min.	Gage
204. Diameter of groove in firing pin cavity, min.	Gage
205. Diameter of large cavity, min.	Gage
206. Diameter of body	Gage
207. Diameter of pinhole on body	Gage
208. Depth to base of large cavity	Gage
209. Depth of gas porthole	Gage
210. Depth to base of firing pin cavity	Gage
211. Depth of drilled cavity	Gage
212. Depth of drilled cavity, max.	Gage
213. Length of body	Gage
214. Width of rectangular section	Gage
215. Width of large groove, min.	Gage
216. Width of slot in head	Gage

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	Defect	Method of inspection
217.	Thickness of rectangular section	Gage
218.	Location of gas port from mounting hole	Gage
219.	Location of setscrew hole	Gage
220.	Location of large groove	Gage
221.	Location of pinhole on body	Gage
222.	Location of mounting hole	Gage
223.	Diameter of drilled cavity	Measure
224.	Angle of pinhole	Measure
225.	Length of slot	Measure
226.	Length of slot relief	Measure
227.	Width of groove in firing pin cavity	Measure
228.	Location of gas port	Measure
229.	Location of slot	Measure
230.	Location of groove in firing pin cavity	Measure
231.	Permissible variation of groove sidewalls, max.	Measure
232.	Location of marking, max.	Measure
233.	Profile of mounting end	Measure
234.	Radius or chamfer missing or incorrect	Visual
235.	Marking missing, incorrect or illegible	Visual
236.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.20 *Tube, Motor, Dwg. F8596917, covering a detail of Dwg. F8596883.*

	Defect	Method of inspection
CRITICAL:		
	None defined.	
MAJOR:		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Concentricity of internal groove with counterbore	Gage
104.	Concentricity of external groove with small body	Gage
105.	Concentricity of small body with counterbore	Gage
106.	Symmetry of pinhole with groove	Gage
107.	Symmetry of width of slot with groove	Gage
108.	True position of thread with counterbore	Gage
109.	True position of hole at extreme end of tube with small body and threaded hole	Gage
110.	Diameter, inside	Gage
111.	Diameter of external groove	Gage
112.	Diameter of internal groove	Gage
113.	Diameter of large body, min.	Gage
114.	Diameter of counterbore, both end, max.	Gage
115.	Diameter of hole at extreme end of tube, max.	Gage
116.	Diameter of small body, min.	Gage
117.	Width of external groove, max.	Gage
118.	Width of internal groove, max.	Gage
119.	Thickness of wall, min.	Gage
120.	Straightness	Gage
121.	Metal defective	Visual
122.	Protective finish inadequate or defective	Visual
123.	Burr or sharp corner	Visual
124.	Foreign matter	Visual
MINOR:		
201.	Pitch diameter of thread, min.	Gage
202.	Diameter of large body, max.	Gage
203.	Diameter of small body, max.	Gage
204.	Diameter of hole at extreme end of tube, min.	Gage
205.	Diameter of pinhole in groove	Gage
206.	Diameter of counterbore, both ends, min.	Gage

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Defect	Method of inspection
207. Total length	Gage
208. Width of external groove, min.	Gage
209. Width of internal groove, min.	Gage
210. Width of slot on large body	Gage
211. Location of external groove	Gage
212. Location of internal groove	Gage
213. Location of hole at extreme end of tube	Gage
214. Location of threaded hole	Gage
215. Length of slot on large body	Measure
216. Length of slot relief on large body	Measure
217. Depth of counterbore, small end	Measure
218. Depth of counterbore, large end	Measure
219. Location of marking	Visual
220. Marking missing, incorrect or illegible	Visual
221. Radius or chamfer missing or incorrect	Visual
222. Surface finish, improper	Visual

4.3.2.21 *Tube, Launcher & Pellet Assembly, Dwg. F8597935, covering a detail of Dwg. F8596883 (before insertion of pellet).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of internal thread, max.	Gage
104. Minor diameter of internal thread, max.	Gage
105. Concentricity of pitch diameter of external thread with inside diameter	Gage
106. Concentricity of pitch diameter of internal thread with inside diameter	Gage
107. Concentricity of large diameter with pitch diameter of internal thread	Gage
108. Diameter large body, min. internal threaded end	Gage
109. Diameter of body, min.	Gage
110. Diameter, inside, max.	Gage
111. Diameter of external undercut, min.	Gage
112. Diameter of internal undercut, max.	Gage
113. Thickness of wall, min.	Gage
114. Straightness	Gage
115. Metal defective	Visual
116. Protective finish, inadequate or defective	Visual
117. Surface finish of inside diameter, improper	Visual
118. Burr or sharp corner	Visual
119. Foreign matter	Visual
MINOR:	
201. Pitch diameter of external thread, max.	Gage
202. Pitch diameter of internal thread, min.	Gage
203. Diameter of large body, external threaded end	Gage
204. Diameter of large body, max., internal threaded end	Gage
205. Diameter inside, min.	Gage
206. Length to external undercut	Gage
207. Length to internal undercut	Gage
208. Total length	Gage
209. Radius at inside diameter, external threaded end	Gage
210. Length of large diameter, external threaded end	Measure
211. Length of large diameter, internal threaded end	Measure
212. Width of external undercut	Measure
213. Width of internal undercut	Measure
214. Marking missing, incorrect or illegible	Visual
215. Radius or chamfer missing or incorrect	Visual
216. Surface finish, improper, except as otherwise specified	Visual

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4.3.2.22 *Tube, Launcher & Pellet Assembly, Dwg. F8597935, covering a detail of Dwg. F8596883 (after insertion of pellet).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Protrusion of pellets over thread, max.	Gage
102. Pellet missing, loose or improperly inserted	Manual/Visual
103. Thread damaged	Visual
MINOR:	
201. Location of pellet	Measure

4.3.2.23 *Nozzle (47° 30' Angle) Dwg. F10520244, covering a detail of Dwg. F8596883 (before application of protective finish).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of body (left of flange) with flange	Gage
102. Concentricity of nozzle exit port (at basic diameter) with nozzle throat	Gage
103. Symmetry of slot at nozzle exit port with nozzle throat	Gage
104. True position of nozzle throat with center nozzle throat	Gage
105. True position of center nozzle throat with body	Gage
106. Diameter of body (left of flange), min.	Gage
107. Diameter of nozzle throat	Gage
108. Diameter of step on body, max.	Gage
109. Length of nozzle throat, min.	Gage
110 Location of center port	Gage
111. Location of basic diameter on exit port	Gage
112. Burr or sharp corner	Visual
MINOR:	
201. Diameter of body (left of flange), max.	Gage
202. Diameter of step on body, min.	Gage
203. Length to top of exit side of nozzle face	Gage
204. Location of large flange	Gage
205. Thickness of large flange	Gage
206. Depth of groove in nozzle exit port	Gage
207. Width of groove in nozzle exit port	Gage
208. Length of step on body	Measure

4.3.2.24 *Nozzle (47° 30' Angle) Dwg. F10520244, covering a detail of Dwg. F8596883 (after application of protective finish).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of flange with body	Gage
102. Concentricity of body with large cavity	Gage
103. Concentricity of middle groove with body	Gage
104. Concentricity of groove (right end) with body	Gage
105. Concentricity of bore in elbow with body	Gage
106. Concentricity of groove (left end) with flange	Gage
107. Symmetry of flat bottom hole with middle groove	Gage
108. True position of hole on body with body and center nozzle throat	Gage
109. Diameter of flange, min.	Gage
110. Diameter of body, min.	Gage
111. Diameter of small cavity in elbow, max.	Gage

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Defect	Method of inspection
112. Diameter of nozzle throat (4 lower ports)	Gage
113. Diameter of groove (left end), min.	Gage
114. Diameter of groove (right end), min.	Gage
115. Diameter of middle groove	Gage
116. Diameter of large cavity in elbow, max.	Gage
117. Diameter of hole on body adjacent to flange, max.	Gage
118. Width of middle groove, max.	Gage
119. Depth of hole on body adjacent to flange, max.	Gage
120. Depth of hole in middle groove, max.	Gage
121. Location of large flange from edge of middle groove	Gage
122. Thickness of wall in elbow at large cavity, min.	Gage
123. Thickness of wall connecting elbow, min.	Visual
124. Metal defective	Visual
125. Protective finish inadequate or defective	Visual
126. Burr or sharp corner	Visual
127. Foreign matter	Visual
MINOR:	
201. Diameter of flange, max.	Gage
202. Diameter of body, min.	Gage
203. Diameter of groove (left end), max.	Gage
204. Diameter of groove (right end), max.	Gage
205. Diameter of large cavity in elbow, min.	Gage
206. Diameter of hole in body adjacent to flange, min.	Gage
207. Diameter of small cavity in elbow, min.	Gage
208. Diameter of elbow body	Gage
209. Diameter of hole in middle groove	Gage
210. Diameter of inlet port	Gage
211. Depth of hole on body adjacent to flange, min.	Gage
212. Depth of hole in middle groove, min.	Gage
213. Total length	Gage
214. Length of protrusion of elbow	Gage
215. Thickness of small flange	Gage
216. Location of middle groove	Gage
217. Location of hole in middle groove	Gage
218. Location of hole on body adjacent to flange	Gage
219. Width of groove (left end)	Gage
220. Width of groove (right end)	Gage
221. Thickness between inlet and exit sides of port	Gage
222. Angle of inlet port	Gage
223. Angle between large cavity and inlet face of ports	Measure
224. Depth of large cavity in elbow	Measure
225. Location of top portion of small cavity in elbow	Measure
226. Location of lower portion of small cavity in elbow	Measure
227. Radius of cutout at end of nozzle	Measure
228. Permissible variation of groove sidewalls, max.	Visual
229. Radius or chamfer missing or incorrect, except as classified	Visual
230. Marking, missing or illegible	Visual
231. Surface finish improper	Visual

4.3.2.25 *Cup, Nozzle, Dwg. C8597583, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Thickness of wall, min.	Gage
102. Location of large basic diameter	Gage
103. Location of small basic diameter	Gage
104. Radius of conical surface	Gage

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Defect	Method of inspection
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR:	
201. Total length	Gage
202. Thickness of wall, max.	Gage
203. Surface finish, improper	Visual

4.3.2.26 *Plate, Orifice, Dwg. D8597584, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of cavity with body	Gage
102. Concentricity of flange with body	Gage
103. True position of hole with body	Gage
104. Symmetry of slot with body	Gage
105. Diameter of hole	Gage
106. Diameter of body, min.	Gage
107. Diameter of flange, min.	Gage
108. Width of slot, min.	Gage
109. Thickness of wall at base of small bore	Gage
110. Angle at base of small bore	Gage
111. Metal defective	Visual
112. Protective finish inadequate or defective	Visual
113. Burr or sharp corner	Visual
114. Foreign matter	Visual
MINOR:	
201. Diameter of body, max.	Gage
202. Diameter of large bore	Gage
203. Diameter of small bore	Gage
204. Diameter of flange, max.	Gage
205. Depth of slot	Gage
206. Width of slot, max.	Gage
207. Thickness of flange	Gage
208. Thickness at base of large bore	Gage
209. Total length	Measure
210. Radius or chamfer missing or incorrect	Visual
211. Marking missing, incorrect or illegible	Visual
212. Surface finish, improper	Visual

4.3.2.27 *Collar, Grain, Dwg. C8597664, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Diameter, outside, min.	Gage
104. Diameter inside, max.	Gage
105. Metal defective	Visual
106. Protective finish, inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, min.	Gage
202. Diameter inside, min.	Gage

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Defect	Method of inspection
203. Diameter, outside, max.	Gage
204. Location of hole	Gage
205. Diameter of countersink	Measure
206. Total length	Measure
207. Chamfer missing or incorrect	Visual
208. Marking missing, incorrect or illegible	Visual
209. Surface finish, improper	Visual

4.3.2.28 *Can, Bottom, Dwg. F8597937, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of small body with body	Gage
104. Concentricity of groove with body	Gage
105. Symmetry of boss with small cavity	Gage
106. True position of thread with body	Gage
107. Diameter of small body, min.	Gage
108. Diameter of groove, min.	Gage
109. Diameter of body, min.	Gage
110. Length of effective thread, min.	Gage
111. Length of imperfect thread, max.	Gage
112. Depth of large cavity, max.	Gage
113. Depth of small cavity, max.	Gage
114. Thickness of large cavity wall, min.	Gage
115. Thickness of small cavity wall, min.	Gage
116. Metal defective	Visual
117. Protective finish inadequate or defective	Visual
118. Burr or sharp corner	Visual
119. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, min.	Gage
202. Diameter of body, max.	Gage
203. Diameter of small body, max.	Gage
204. Diameter of small cavity	Gage
205. Diameter of large cavity	Gage
206. Diameter of groove, max.	Gage
207. Depth to boss	Gage
208. Depth of small cavity, min.	Gage
209. Depth of large cavity, min.	Gage
210. Total length	Gage
211. Length of small body	Gage
212. Depth of slot on boss	Gage
213. Width of groove	Measure
214. Width of boss	Measure
215. Width of slot on boss	Measure
216. Location of slot on boss	Measure
217. Location of groove on body	Measure
218. Permissible variation of groove sidewall	Measure
219. Marking missing, incorrect or illegible	Visual
220. Radius or chamfer missing or incorrect	Visual
221. Surface finish, improper	Visual

4.3.2.29 *Trunnion and Pellet Assembly, Dwg. F8597936, covering a detail of Dwg. F8596883 (before insertion of pellet).*

Defect	Method of inspection
CRITICAL:	
None defined.	

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Defect	Method of inspection
MAJOR:	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with bore	Gage
104. Concentricity of lug	Gage
105. Symmetry of width across base of lugs with bore	Gage
106. True position of lug with bore	Gage
107. Diameter of undercut, min.	Gage
108. Diameter of lug	Gage
109. Diameter of bore, max.	Gage
110. Width across base of lugs	Gage
111. Wall thickness, min.	Gage
112. Metal defective	Visual
113. Protective finish inadequate or defective	Visual
114. Surface finish, improper	Visual
115. Burr or sharp corner	Visual
116. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, max.	Gage
202. Minor diameter of thread, max.	Gage
203. Diameter of body	Gage
204. Diameter of bore, min.	Gage
205. Diameter of pinhole in eyelet	Gage
206. Length to undercut	Gage
207. Total length	Gage
208. Location of lug	Gage
209. Location of pinhole in eyelet	Measure
210. Location of eyelet	Measure
211. Location of boss at base of lug	Measure
212. Length of lug	Measure
213. Length of eyelet	Measure
214. Width of boss at base of lug	Measure
215. Width of undercut	Measure
216. Width of eyelet, max.	Measure
217. Width at top of eyelet	Measure
218. Radius or chamfer missing or incorrect	Visual
219. Marking missing, incorrect or illegible	Visual
220. Surface finish, improper, except as otherwise specified	Visual

4.3.2.30 *Trunnion and Pellet Assembly, Dwg. F8597936, covering a detail of Dwg. F8596883 (after insertion of pellet).*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Protrusion of pellet over thread, max.	Gage
102. Pellet, missing, loose or improperly inserted	Manual/Visual
103. Thread damaged	Visual
MINOR:	
201. Location of pellet	Measure

4.3.2.31 *Cylinder, Slider Valve and Booster Tube Assembly, Dwg. F8596893, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Concentricity of tube and with cylinder head	Gage
102. Diameter of body at brazed joint	Gage

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Defect	Method of inspection
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR:	
201. Total length	Gage
202. Marking missing, incorrect or illegible	Visual

4.3.2.32 *Nozzle, Assembly, Dwg. F8596905, covering a detail of Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Symmetry of tang slot with hole in nozzle body	Gage
102. Diameter of tang body at brazed joint	Gage
103. Tang and retainer improperly brazed	Visual
104. Metal defective	Visual
105. Protective finish, inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR:	
201. Total length	Gage
202. Cup protruding above groove in nozzle	Visual
203. Marking missing, incorrect or illegible	Visual

4.3.2.33 *Catapult, Aircraft Ejection Seat, M9, Metal Parts Assembly, Dwg. F8596883.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Any part missing, except as otherwise classified	Visual
102. Improper assembly	Visual
103. O-ring not lubricated	Visual
104. Marking missing, incorrect or illegible	Visual
MINOR:	
201. Fungus or moisture resistant varnish missing	Visual
202. Shipping plug missing	Visual
203. Protective finish inadequate or defective	Visual

4.3.2.34 *Container, Ammunition, Fiber, Dwg. C8826679.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
None defined.	
MINOR:	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape, incorrect	Visual

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4.3.2.35 *Box, Packing, Ammunition, Dwg. C8826680.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
None defined.	
MINOR:	
201. Contents loose	Manual
202. Hardware missing, loose, braken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4, except as provided in Specification MIL–I–45208.

4.3.3.1 *Hardness.* Fifty units from each heat-treated lot of components shall be tested for compliance with the requirements of the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cau e for rejection of the lot of components.

4.3.3.2 *Hydrostatic pressure.* Each component listed in 3.5 shall be subjected to an internal hydrostatic pressure. Failure of any component to comply with the requirements of 3.5 shall be cause for rejection of the component.

4.3.3.2.1 *Breech launcher.* Each breech launcher shall be subjected to a hydrostatic pressure. Failure of any component to comply with the requirements of 3.5.1 shall be cause for rejection of the component.

4.3.3.3 *Air pressure.* Each nozzle assembly shall be subjected to an air pressure test. Failure of any assembly to comply with the requirements of 3.6 shall be cause for rejection of the assembly.

4.3.3.4 *Tensile load.* Each nozzle assembly shall be subjected to the tensile load test. Failure of any assembly to comply with the requirements of 3.7 shall be cause for rejection of the assembly.

4.3.3.5 *Magnetic particle and penetrant.* The outside surface of each motor tube shall be subjected to a magnetic particle or penetrant inspection to determine presence of cracks. Failure of any unit to comply with requirements of 3.13.1 and Specifications MIL–I–6866 or MIL–I–6868 as applicable shall be cause for rejection of the unit.

4.3.3.6 *Shearpin strength.* Fifty pins from each lot of shearpins shall be tested for shear strength. Failure of any pin to comply with the requirements of 3.8 shall be cause for rejection of the lot of shearpins.

4.3.3.7 *Valve shearpin strength.* Fifty pins from each lot of valve shearpins shall be tested for shear strength. Failure of any pin to comply with the requirements of 3.9 shall be cause for rejection of the lot of valve shearpins.

4.3.3.8 *Structural strength.* Fifteen catapults from each lot shall be subjected to the structural strength test. Failure of any unit to comply with the requirements of 3.10 shall be cause for rejection of the lot of catapults.

4.3.3.9 *Protective finish.* The number of specimens and acceptance criteria for protective finish test shall be in accordance with the applicable military standard listed on the drawing.

4.3.3.10 *Residual magnetism.* Each catapult shall be tested for excessive residual magnetism. Failure of any unit to comply with 3.12 shall be cause for rejection of the catapult.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment

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listed on IEL–8596883, except as provided in Specification MIL–I–45208.

4.4 **Test methods and procedures.**

4.4.1 *Hardness.* Prior to the application of protective finish, a hardness testing machine shall be used to determine compliance with the requirements specified on the detail drawing.

4.4.2 *Hydrostatic pressure.* Prior to the application of protective finish, each component listed in 3.5 and 3.5.1 shall be individually mounted in the device, using the fixtures for sealing all openings and subjected to the hydrostatic pressure to determine compliance with the requirements listed in 3.5 and 3.5.1. The breech launcher shall be sealed such that the pressure shall be applied to an area 1¾ inches, minimum, from the open end of the breech and transmit no load to the bottom surface.

4.4.3 *Air pressure.* The cup in the nozzle assembly shall be subjected to the air pressure test. The nozzle assembly shall be so placed in the fixture as to seal all openings before the air pressure is applied so that each cup will be subjected equally to the required pressure.

4.4.4 *Tensile load.* The load designated in 3.7 shall be applied and shall register on the dial indicator calibrated to the applied load.

4.4.5 *Shearpin strength.* The test shall be performed after application of protective finish. The pin shall be inserted in the fixture and tested in conjunction with a Tensile Testing Machine. The load shall be applied at the rate of 0.125 inches per minute.

4.4.6 *Valve shearpin strength.* The test shall be performed after application of protective finish. The pin shall be inserted in the fixture and tested in conjunction with a Tensile Testing Machine. The load shall be applied at the rate of 0.125 inches per minute.

4.4.7 *Magnetic particle and penetrant.* The test shall be performed prior to the application of protective finish in accordance with the procedures established in Specifications MIL–I–6866 or MIL–I–6868, as applicable, to determine compliance with the requirements of 3.13.1.

4.4.8 *Structural strength.* The catapult shall be secured between the mounting point and the load in tension, and compression applied to determine compliance with the requirements of 3.10.

4.4.9 *Protective finish.* Test procedures for protective finish shall be in accordance with the applicable military standard referenced on the drawing.

4.4.10 *Residual magnetism.* The catapult shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the catapult 6 inches from, and at the same height as, the compass in a north-south horizontal position with the compass set in an east-west heading as shown in figure 1. The catapult shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180° in the horizontal position and moved slowly back past the compass.

5. PREPARATION FOR DELIVERY

5.1 **Packing and marking.** (Level A.) Packing and marking shall be in accordance with Drawings C8826679 and C8826680.

5.2 **Data cards.** Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:
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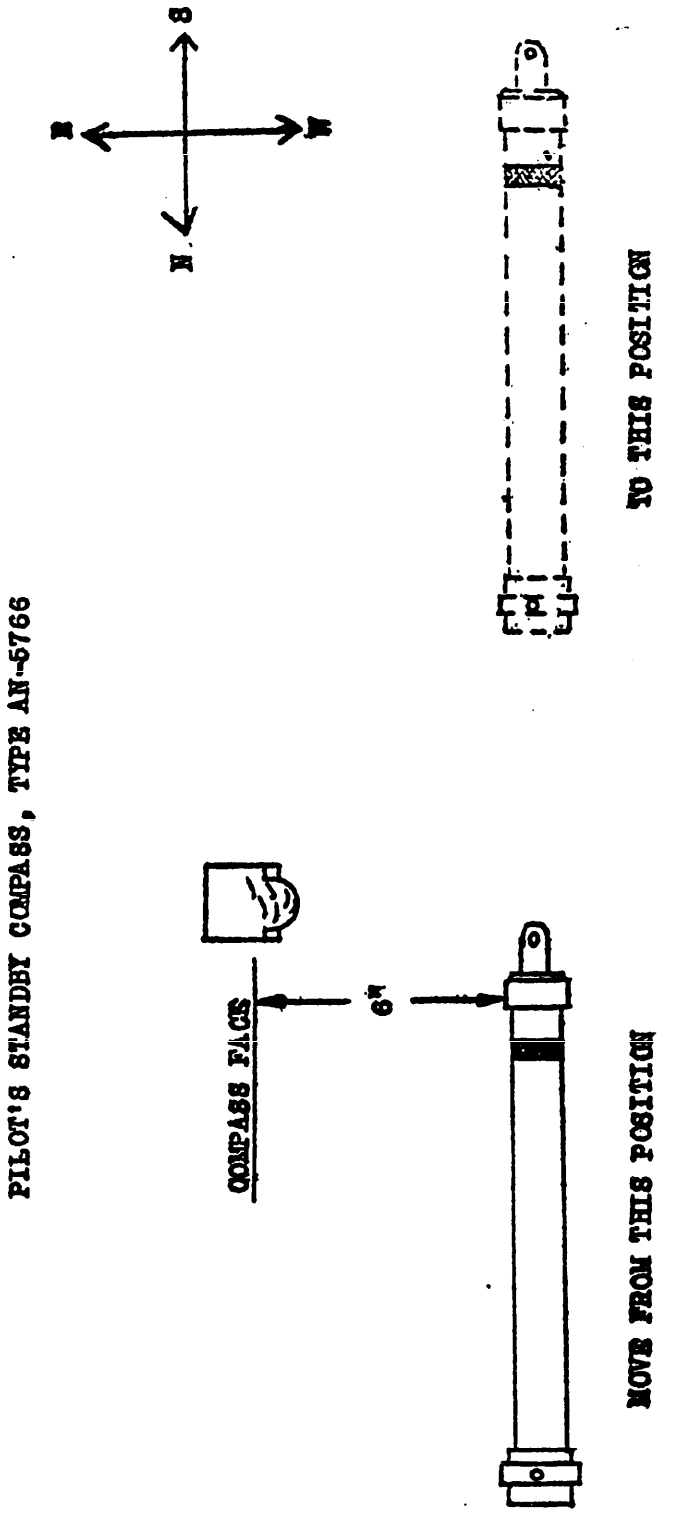


FIGURE 1. *Layout for determination of residual magnetism in catapult, aircraft ejection seat, M9.*

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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